



Fe/polymer joining via Fe/TiB₂ composite structures via in-situ laser-induced reaction of Fe-Ti-B system: Effect of powder composition

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ARTICLE INFO

Keywords:

Metal/polymer joining
Mechanical behavior
Laser processing
Fe-Ti-B system
Joining strength

ABSTRACT

Achieving strong direct joining between steel and polymers through mechanical interlocking is crucial for developing multi-material structures, particularly in the automotive and aerospace industries. This study synthesized micro-scale structures on a pure Fe substrate (simulating interstitial-free (IF) steel) for mechanical interlocking with thermoplastic parts. Numerous submillimeter-scale Fe/TiB₂ composite particles were in-situ synthesized by laser scanning on the Fe-Ti-B powder mixture and well-bonded with the Fe substrate. The effects of powder composition (TiB₂ volume fraction) on the morphology, microstructure, and joint strength with PA6 were investigated. A TiB₂ volume fraction over 60 % was essential for the formation of the composite particles promoted by a TiB₂ skeletal structure. Higher TiB₂ volume fractions increased the area fraction of the composite particles and decreased the bonding ratio (adhesive) of the particles with the substrate due to poor adhesiveness at the edge of the laser-scanning line. A wider high-temperature region was generated at a higher TiB₂ volume fraction, suggesting that the reaction heat to form TiB₂ assisted the bonding of the particles with the substrate at the edge of the laser scanning line. The Fe/PA6 joint strength increased to approximately 30 MPa with increasing the TiB₂ volume fraction to 100 % and showed a linear correlation with the product of particle area fraction and bonding ratio. A higher TiB₂ volume fraction was preferable for enhancing the joint strength via the micro-scale structures synthesized by laser scanning on the Fe-Ti-B powder mixture. A combination of the micro-structuring process using a high fraction of TiB₂ with advanced joining technologies will contribute to manufacturing high-strength Fe/polymer hybrid parts.

Introduction

Lightweight designs are crucial for modern industries aiming to achieve carbon neutrality (Wei et al., 2024). Chueh et al. (2020) reviewed the prerequisites for lightweight design, which necessitates the integration of multiple materials with distinct properties. These materials are tailored to specific areas within highly functional devices or components through a streamlined process.

The joining process of lightweight dissimilar materials is crucial in various engineering applications, particularly in the automobile, aerospace, medical, and biological industries (Nazir et al., 2023; Zhou et al., 2022). The most widely investigated category for hybrid structures is metal-to-polymer joints (Siena et al., 2023; Sun et al., 2021). Among metals, the majority of research has been conducted on titanium, copper (Chen et al., 2024), aluminum, and steels to enhance characteristics such as hardness, conductivity, magnetic properties, wear resistance, strength, and thermal performance (Jiao et al., 2021; Sun et al., 2022a,

2022b). For polymers, materials like PA6, PEEK, and PET are commonly studied for their physicochemical resistance and lightweight properties (Feng et al., 2018).

However, the differences in mechanical and physical characteristics between metal and polymer present significant challenges in the joining process. Mechanical joining (through rivets or bolts) (Ravichandran et al., 2024), adhesive joining, and direct joining have been successfully proven as unification methods in the evolution of hybrid structures. Mechanical joining employs methods such as bolts and rivets for mechanical fastening, offering benefits like minimal surface treatment, reliable bonding, and the potential for easy disassembly (Vasconcelos et al., 2023). Nevertheless, it faces challenges such as stress concentration at the joining points, increased costs, and diminished weight-saving effects due to the increase in the number of parts (Vasconcelos et al., 2023). Kleffel et al. (2017) note that adhesive joining enables full-faced bonding without stress concentrations, which also has the advantages of vibration reduction and corrosion prevention (sealed joint). However,

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<https://doi.org/10.1016/j.jajp.2024.100249>

concerns regarding the time-consuming nature of the process and degradation of adhesives in high-temperature and high-humidity environments also exist.

The direct joining method primarily relies on mechanical interlocking and/or chemical bonding (Wu et al., 2020). Artificial microstructures (e.g., holes, protrusions, dimples, and grooves) formed on the metal surface facilitate micro-scale mechanical interlocking and enhance joint strength. A key requirement for this process is the infiltration of the polymer into these microstructures during the injection molding process. Since the adherents are joined directly without mechanical fasteners or adhesives, direct joining can avoid significant stress concentration and environmental degradation associated with the aforementioned methods.

Various methods have been proposed to create dimples on metal surfaces, including dry and wet processing techniques (Enami et al., 2020). In dry processing, laser machining creates surface structures on the scale of several dozen micrometers (Taki et al., 2016), and sand-blasting creates structures on the scale of several micrometers (Kajihara et al., 2018). Wet processing methods, such as chemical etching, can form nanoscale porous structures to modify the metal surface (Tong et al., 2021; Zhao et al., 2024). These methods present the potential to fabricate metal/polymer hybrid structures. The infiltration of polymer into dimples enhances joint strength through mechanical interlocking, intermolecular forces, and/or chemical bonding, primarily depending on surface roughness and the presence of undercut (Kleffel et al., 2017). Consequently, more intricate topologies can enhance tensile strength perpendicular to the bonding interface, compared to simpler concave-convex structures. Mao et al. proposed improved structural features (concave and moat microstructures) to alleviate the adhesive protrusion problem at the bonding interface, while also extending the bondable thickness range of SU-8 adhesives (Mao et al., 2019a, 2019b).

Recently, Suzuki et al. (2018a, 2018b) proposed a direct joining process for metals and polymers via an “interpenetrating phase layer”. An open-cell porous-structured layer is formed on the metal substrate through a space holder method. They proved that such intricate morphology helps alleviate anisotropy and achieve strong interlocking after polymer infiltration. Meanwhile, the composite layer formed between the metal and polymer reduces the adverse effects of heat.

Kim et al. (2019) proposed a more practical processing method to produce the composite layer by powder sintering without applying pressure at high temperatures. Simultaneously, the in-situ technique of local heating by laser effectively minimizes the thermal effect on the substrate, leading to improved adhesion at interfaces and, consequently, enhanced mechanical properties. By applying in-situ/one-step laser scanning to an Al-Ti-C powder mixture, Ti and C react to form TiC, releasing a large amount of reaction heat. Notably, the synthesized TiC/aluminum composite forms a micro-scale particulate structure that is rooted on an aluminum plate. Unlike dimples, stress concentration is mainly present on the metal side, and the formation of microscopic undercuts along the surface of the metal leads to increased strength both in shear and tension (Geng et al., 2023). In this context, analyzing the morphology and adhesiveness of microstructures is crucial, as they directly influence joint strength (Huang et al., 2022).

Previous studies were primarily dedicated to Al-SiC systems. However, various types of steel (ferrous materials) are currently used in a major part of automotive body structures. In industrial applications, ferrous-based matrices have attracted more attention due to their low cost and excellent mechanical properties, offering higher strength compared to aluminum (Aparicio-Fernández et al., 2016). Wang et al. (2014) analyzed the microstructural features and distribution of the composites with the introduction of TiB₂ particles, and simultaneously proved enhanced mechanical properties with increased TiB₂ content. Furthermore, TiB₂ has been selected for its thermodynamic stability within the Fe matrix, making it one of the optimal strengthening phases for iron (Feng et al., 2023). Its high Young's modulus ($E = 565$ GPa) and low density ($4.5 \text{ g}\cdot\text{cm}^{-3}$) are expected to promote strong interfacial

bonding (Feng et al., 2023). More importantly, TiB₂ can be synthesized in-situ from Fe-Ti-B powder agglomerates through a pseudo-binary eutectic reaction, contributing favorably to the material's good mechanical properties and damage resistance (Suzuki et al., 2019). Meanwhile, the effects of particle size and blending ratio on the porous structure and bonding quality have already been investigated.

The purpose of this study was to fabricate micro-scale structures on a pure Fe substrate (simulating interstitial-free (IF) steel sheets used for automotive body panels) via in-situ laser-induced reaction within the Fe-Ti-B system. Concurrently, to promote Fe/polymer joining, the effects of powder composition on the morphology, microstructure, and joint strength of these micro-scale structures were also explored.

Materials and methods

Powders and substrates

Fig. 1 shows scanning electron microscopy (SEM) images of the raw powders used in this study. Fe powder (99.99 % purity, particle size $< 53 \mu\text{m}$, Koujundo Chemical Lab. Co., Ltd.), Ti powder (99.9 % purity, particle size $< 45 \mu\text{m}$, Koujundo Chemical Lab. Co., Ltd.) and B powder (99 % purity, particle size $< 45 \mu\text{m}$, Koujundo Chemical Lab. Co., Ltd.) were used to prepare starting powders. These powders were dry blended for 30 min. The molar ratio of Ti and B was fixed at 1:2 to synthesize TiB₂.

Based on the assumption that all Ti and B in the starting powder mixture would react and convert into TiB₂, the volume fractions of TiB₂ were varied from 0 %, 40 %, 60 %, 80 %, to 100 %.

Based on thermodynamic calculations using the CALPHAD method with a thermodynamic database for an Fe-based multicomponent system (PanFe), the liquidus projection of the Fe-Ti-B ternary system was calculated under equilibrium solidification conditions (Fig. 2) (Chang et al., 2004). It encompasses the overall composition of the studied Fe-Ti-B powder mixtures, especially where the molar ratio of Ti to B equals 0.5, as presented in a dotted line (Zhang et al., 2016).

Prior to laser irradiation, the as-received pure Fe substrate (purity > 99.95 %, $20 \text{ mm} \times 50 \text{ mm} \times 1 \text{ mm}$) was polished with SiC papers (#120, #320) to remove its oxide film and keep the surface roughness constant. Subsequently, as shown in Fig. 3, the powder mixture was stacked in a rectangle form with dimensions of $18 \text{ mm} \times 5 \text{ mm} \times 100 \mu\text{m}$ (length $\times$ width $\times$ height) on the Fe substrate. A pulsed diode laser beam (wavelength: 1070 nm, pulse frequency rate: 1000 Hz, pulse length: 700 μs) was used for irradiation onto the bedded powder. Based on preliminary experiments optimizing energy input and processing speed, the laser power and scanning speed were set at 350 W and 60 mm/s, respectively. The laser scanning length was set to 10 mm. After laser irradiation, the Fe substrate was cleaned by an ultrasonic wave in ethanol for 300 s to remove unreacted powders remaining around the formed micro-scale structure.

A scanning electron microscope (SEM, JSM-6610A, JEOL, Japan) was used to examine the surface morphology and microstructure of the cross-sectioned micro-scale structures. Specimens were cut perpendicular to the laser scanning direction and embedded in epoxy resin. Subsequently, the surfaces were polished using SiC abrasive papers (#320, #800, #1000) and buffed with diamond suspensions (3 μm and 1 μm), followed by a 0.04 μm colloidal silica suspension. The cross-sectional microstructure was observed using the SEM, and elemental distributions were analyzed by energy-dispersive X-ray spectroscopy (EDS).

The obtained topographical and cross-sectional morphology SEM images were quantitatively analyzed using ImageJ, an open-source software. The entire area of the micro-shaped structure was calculated based on the overall micro-structure depicted in Fig. 3(a). The average area fraction of particle-shaped products was calculated multiple times (Fig. 3b and c). The bonding ratio between particle-shaped products and the Fe surface was quantified using the following equation, based on cross-sectional SEM images shown in Fig. 3(d) (Kim et al., 2021).

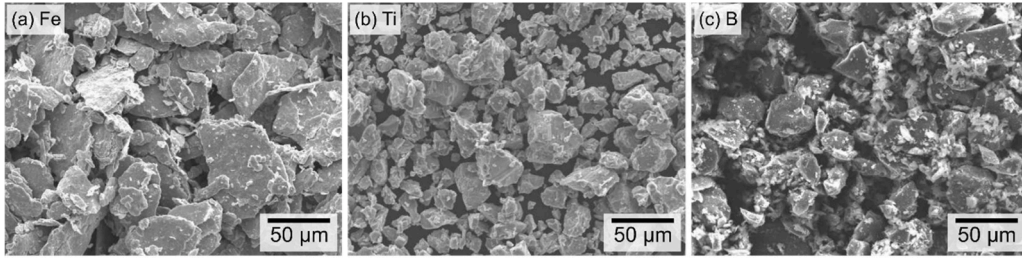


Fig. 1. Scanning electron microscopy (SEM) images showing the starting powders of (a) Fe (size < 53 μm), (b) Ti (size < 45 μm), and (c) B (size < 45 μm) used in the present study.

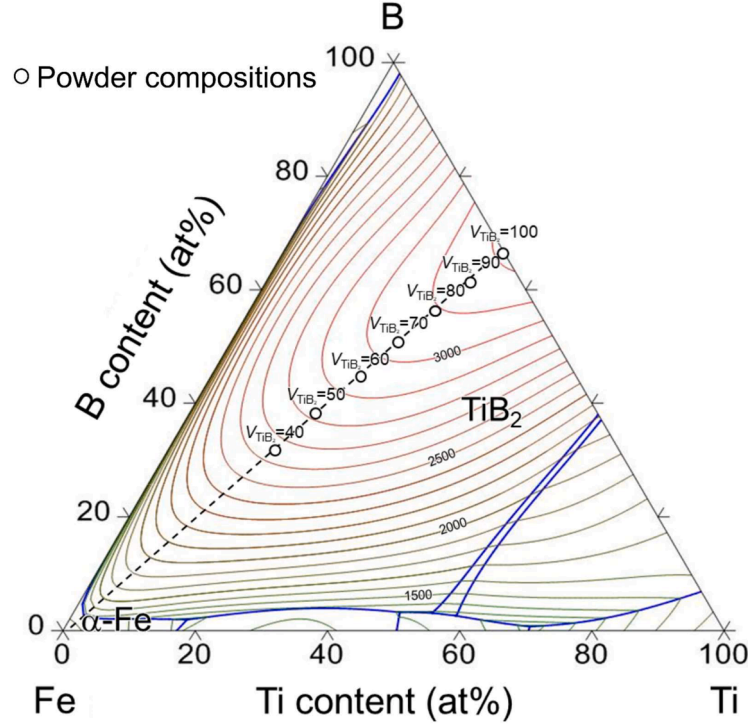


Fig. 2. Liquidus projection of the equilibrium phase diagram for Fe-Ti-B ternary system at Ti:B = 1:2 obtained by the CALPHAD approach.

$$\text{Bonding ratio} = \frac{l_1 + l_2 + l_3 + l_4 + \dots}{L + L_2 + L_3 + \dots} = \frac{\sum_k l_k}{\sum_j L_j} \quad (1)$$

Where l_k and L_j represent the length and width of the bonded part of a particle-shaped structure with the Fe substrate. To ensure statistical significance, more than 30 particle-shaped products were analyzed to calculate the bonding ratio.

X-ray diffraction (XRD) measurements with Cu-K α radiation (Ultima-IV, Rigaku, Japan) were conducted at 40 kV and 40 mA to identify the phases constituting the synthesized micro-scale structures (Wang et al., 2022). To observe the morphology of TiB₂ within the micro-scale structures, etching was carried out with hydrochloric acid for 2 h, consequently dissolving the Fe matrix and leaving behind the synthesized products of the Fe-Ti-B combustion reaction.

Subsequently, as illustrated in Fig. 4, a pre-molded polyamide-6 substrate (PA6, 20 mm × 50 mm × 3 mm, Toray Plastic Precision Co., Ltd.) was used for joining with the pretreated Fe substrate. The overlap between the two specimens was set at 10 mm. A hydraulic hot-pressing machine (AH-10TD, AS ONE Co., Ltd., Japan) was used to fabricate Fe/PA6 joints. The Fe alloy substrates with micro-scale structures were placed on the hot plate of the hydraulic hot press. The joining process was conducted based on the optimized temperature conditions proposed in the preliminary study by Iwata et al. (2022), Kim et al. (2020).

After joint specimens were fabricated, lap shear tests were performed using a universal tensile tester (AG-1 20 kN, Shimadzu Corporation, Japan) to measure joint strength. Fig. 5 shows a schematic illustration of an Fe/PA6 joint for lap shear testing. The joint specimens were loaded at a crosshead speed of 1 mm/min at a temperature of 25 °C. To estimate the ultimate lap shear force, three samples for each type of Fe/PA6 joint were tested. The fracture surfaces after the lap shear tests were characterized by the SEM. The joining area for each type of powder mixture was calculated based on the top view SEM image of each micro-scale structure and image analysis software.

Results

Micro-scale structures for joining

Fig. 6 presents SEM images showing the topographical morphologies of micro-scale structures fabricated from powder mixtures with the TiB₂ volume fraction of (a) 0 %, (b) 40 %, (c) 60 %, (d) 80 %, and (e) 100 %. When the TiB₂ volume fraction was 0 %, a dense layer was formed on the surface (Fig. 5a). At the TiB₂ volume fraction of 40 %, a flat morphology with a smooth surface was formed at the center of the laser scanning line (Fig. 6b). Particle-shaped products were formed far from the centerline.

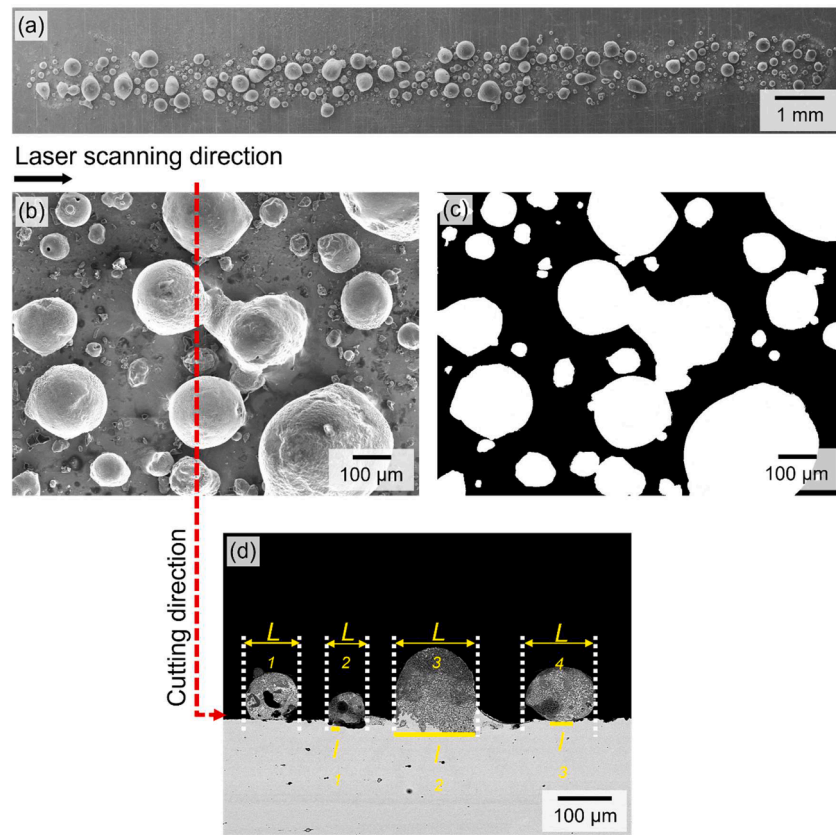


Fig. 3. Representative images used for quantitative analysis: (a) typical entire topographical micro-structure morphologies for area measurement, (b) high-magnification SEM image and (c) its binarized images of the microstructure; (d) cross-sectional SEM image of micro-shaped structure for bonding ratio measurement.

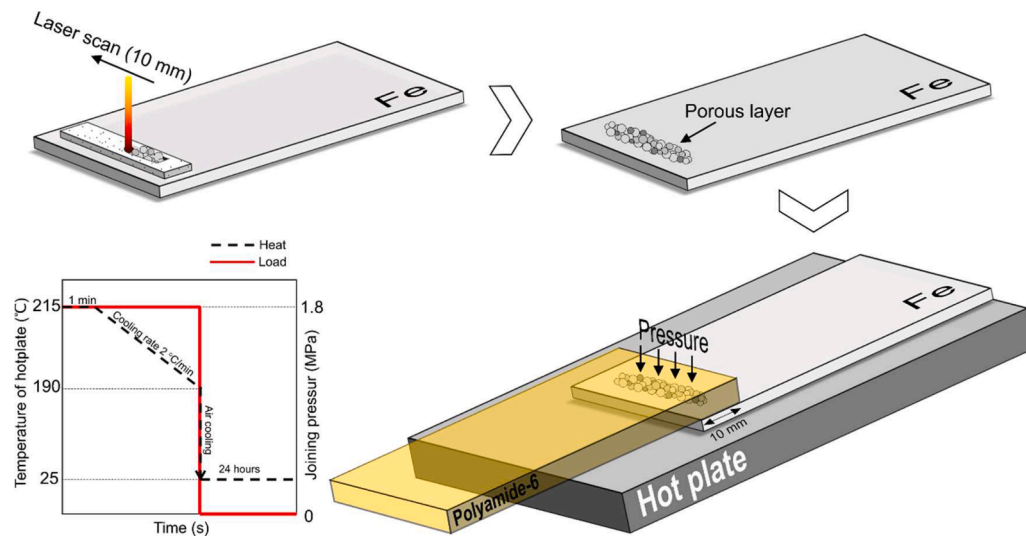


Fig. 4. Schematic illustration showing micro-scale structure formation process on the Fe substrate and Fe/PA6 joint fabrication process, with the history of the temperature of the sample stage and joining pressure.

As the TiB_2 volume fraction increased to 60 %, numerous particles with diameters ranging from tens to hundreds of micrometers were uniformly distributed on the laser scanning line (Fig. 6c). The particles exhibit a predominantly spherical morphology, indicating the threshold for particle-shaped products formation between 40 % and 60 % TiB_2 volume fraction. At TiB_2 fractions exceeding 80 %, particle morphology transitioned from spherical to more intricate, with a significant increase in particle quantity (Fig. 6d and e). In addition, the particles'

distribution was observed to extend perpendicular to the laser scanning direction.

Fig. 7 presents SEM images showing the cross-sections of micro-scale structures fabricated from powder mixtures with TiB_2 volume fractions of (a) 40 %, (b) 60 %, (c) 80 %, and (d) 100 %. At $V_{\text{TiB}_2} = 40$ %, the structure was a flattened shape and well bonded with the Fe substrate (Fig. 7a). When the V_{TiB_2} was 60 %, the structure became more spherical but maintained good adhesiveness with the substrate (Fig. 7b). When

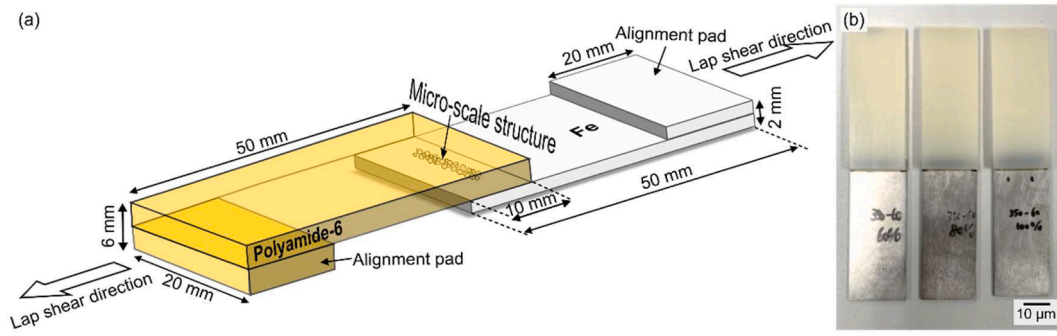


Fig. 5. (a) Schematic illustration and (b) photographs of the Fe/PA6 lap joints used for the lap shear tests.

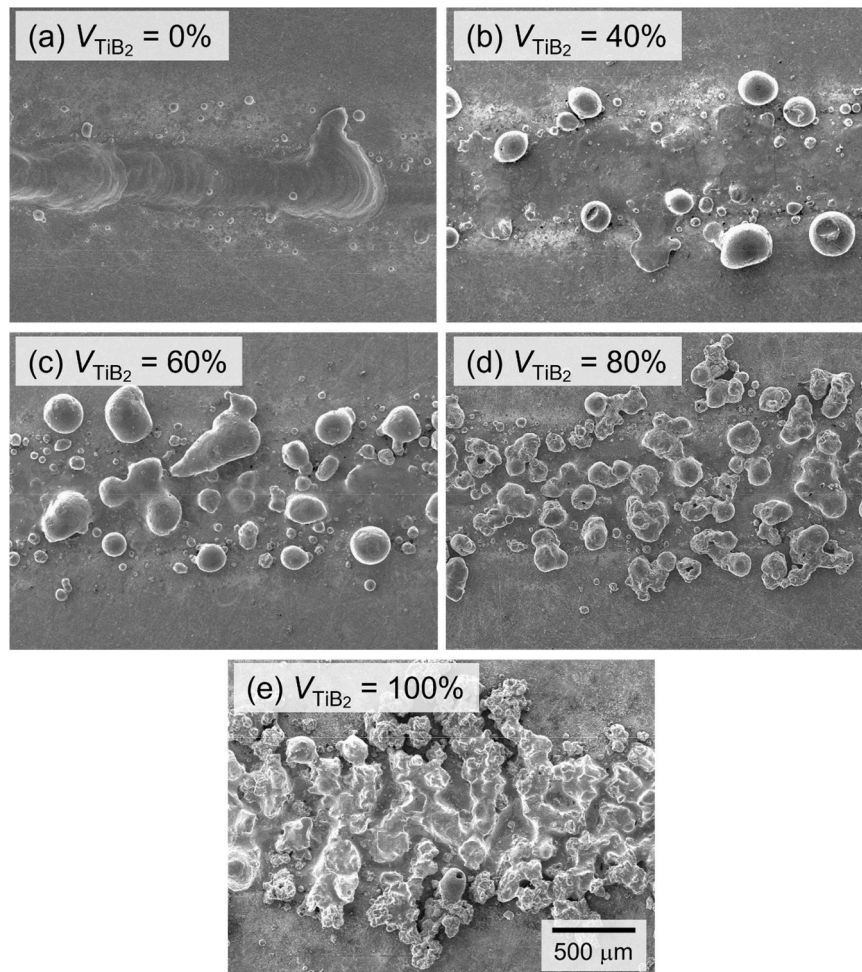


Fig. 6. SEM images showing topographical morphologies of micro-scale structure fabricated from powder mixtures with TiB_2 volume fraction of (a) 0 %, (b) 40 %, (c) 60 %, (d) 80 %, and (e) 100 %.

the V_{TiB_2} exceeded 80 %, the adhesiveness between the particle-shaped products at the edge of the laser-scanning line and the substrate became poor (indicated by arrows), whereas the product at the center maintained good adhesiveness (Fig. 7c and d).

Fig. 8 shows changes in (a) the area fraction of particle-shaped products, (b) the bonding ratio, (c) the product of the area fraction and the bonding ratio, and (d) the area of micro-scale structure as a function of the volume fraction of TiB_2 . The area fraction increased almost linearly with increasing TiB_2 volume fraction (Fig. 8a). This result suggested that the addition of Ti and B powder would promote the formation of the particle-shaped products. However, the bonding ratio

showed a slight decreasing trend at higher TiB_2 volume fraction (Fig. 8b), which may be attributed to the poor adhesiveness of the particle-shaped products far from the center of the laser-scanning line. Despite this, the product of the area fraction and bonding ratio, which presumably correlates with the joint strength with polymer, increased monotonically with the volume fraction of TiB_2 (Fig. 8c), due to the linear increasing trend of the area fraction and the slight decreasing trend of the bonding ratio. The overall area of the micro-scale structure also increased with an increase in TiB_2 volume fraction (Fig. 8d), which corresponded to the extension of the structure perpendicular to the laser scanning direction.

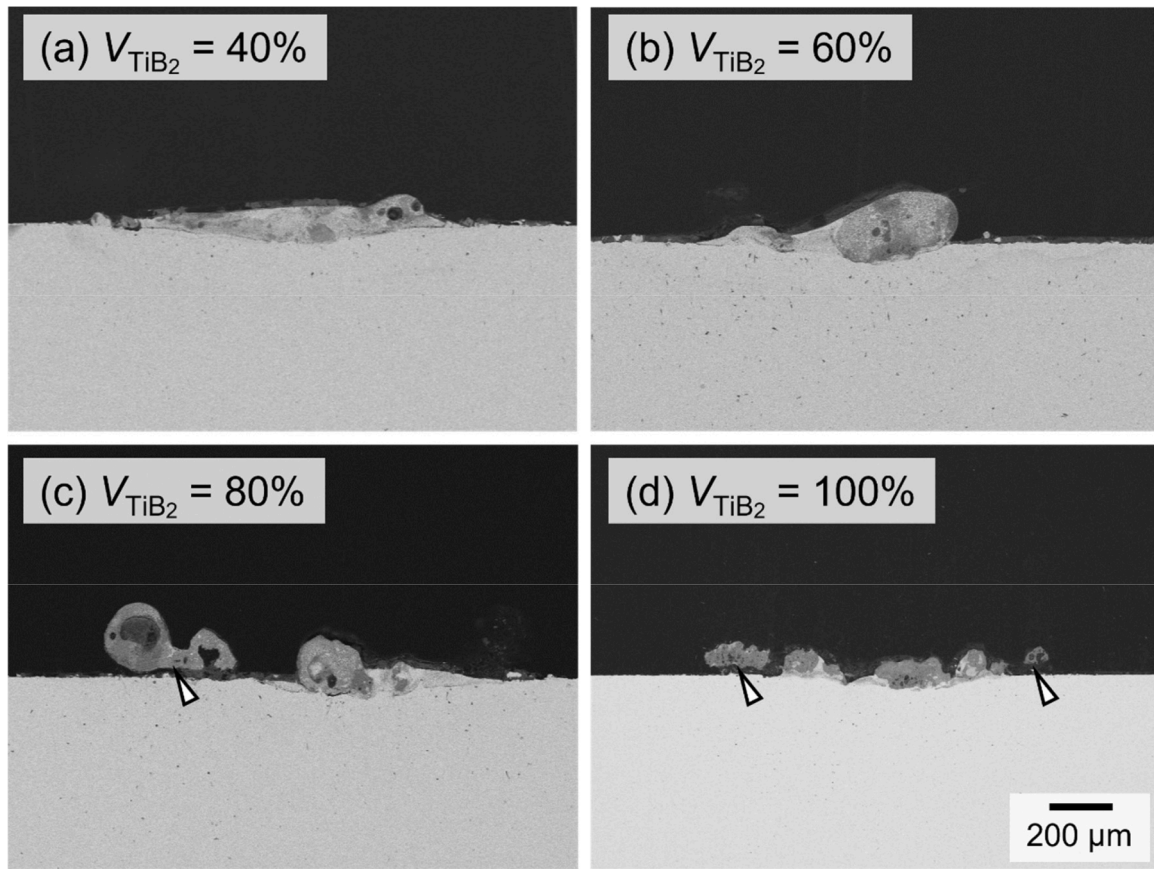


Fig. 7. SEM images showing the cross-sections of micro-scale structures fabricated from powder mixtures with TiB_2 volume fractions of (a) 40 %, (b) 60 %, (c) 80 %, and (d) 100 %.

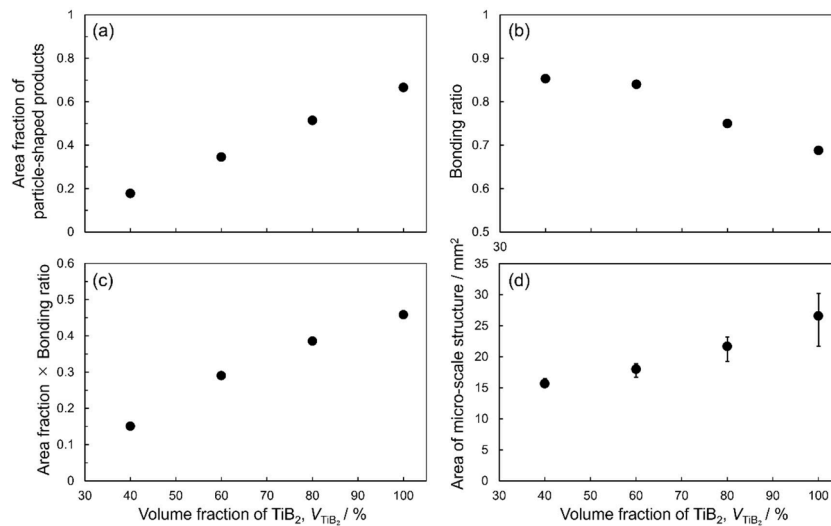


Fig. 8. Changes in (a) the area fraction of particle-shaped products, (b) the bonding ratio, (c) the product of the area fraction and the bonding ratio, and (d) the area of micro-scale structure as a function of the volume fraction of TiB_2 .

Microstructures inside particle-shaped products

Fig. 9 presents XRD profiles for $V_{\text{TiB}_2} = 40\%$, 60% , 80% , and 100% . Across all these conditions, peaks corresponding to $\alpha\text{-Fe}$ and TiB_2 were identified in the XRD profiles, suggesting that the particulate structures are composed of $\alpha\text{-Fe}$ and TiB_2 phases. Moreover, samples fabricated with $V_{\text{TiB}_2} = 100\%$ exhibited more pronounced TiB_2 peaks compared to

the case of $V_{\text{TiB}_2} = 40\%$.

Fig. 10 shows (a) low- and (b) high-magnification SEM images and the corresponding EDS element maps of (c) Fe, (d) Ti, and (e) B showing the microstructure inside the micro-scale structures in the particle-shaped products under $V_{\text{TiB}_2} = 60\%$. Numerous rectangular-shaped dark particles were dispersed in the bright matrix. Fe was highly concentrated in the bright matrix, whereas Ti and B were strongly

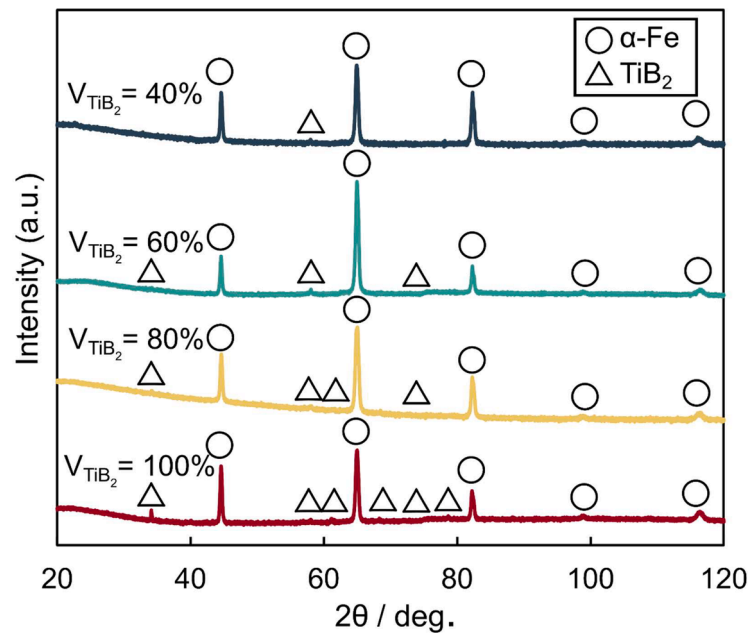


Fig. 9. XRD profiles measured at micro-scale structure fabricated from the powder mixture with TiB_2 volume fraction of 40 %, 60 %, 80 %, and 100 %.

detected inside the dark particles. Based on the XRD profile shown in Fig. 9, the bright matrix and dark particles are identified as $\alpha\text{-Fe}$ and TiB_2 phases, respectively. Thus, elemental Ti and B powders reacted to form TiB_2 during the laser irradiation.

Fig. 11 presents high-magnification SEM images showing the microstructure inside the particle-shaped products fabricated using powder mixtures with $V_{\text{TiB}_2} =$ (a) 40 %, (b) 60 %, (c) 80 %, and (d) 100 %. The actual volume fraction of TiB_2 increased with increasing the designed volume fraction of TiB_2 in the raw powder mixtures. It is noted here that the $\alpha\text{-Fe}$ was observed at the grain boundary of the TiB_2 when the V_{TiB_2} was 100 % (indicated by the arrow and shown by small figures in Fig. 11d). In this condition, Fe was not included in the powder mixture bedded on the substrate. Therefore, the Fe was supplied from the Fe substrate to the particle-shaped products during the laser irradiation.

Characterizations of Fe/PA6 joints

Fig. 12(a) shows load-stroke curves for the Fe/PA6 joints via micro-scale structure fabricated from the powder mixtures with V_{TiB_2} of 60 %, 80 %, and 100 %. The joints basically exhibited linear load-stroke responses and a sharp decrease in the load due to the fracture of the joints although the joints via micro-scale structure fabricated from the powder mixtures with $V_{\text{TiB}_2} = 100$ % exhibited a slight non-linear behavior and wavy curve at the fracture. From the load-stroke curves, the ultimate lap shear strength (N) was quantified and shown in Fig. 12(b). The lap shear strength (N) significantly increased with increasing the TiB_2 volume fraction and the strength of $V_{\text{TiB}_2} = 100$ % was approximately 3 times higher than that of $V_{\text{TiB}_2} = 60$ %. As shown in Fig. 8(c), the area of micro-scale structure increased with increasing the TiB_2 volume fraction. It was hypothesized that the ultimate lap shear strength increased by the enlarged joint area. To consider this possibility, lap shear strength (MPa) was evaluated by dividing the lap shear strength (N) by the area of micro-scale structure. The lap shear strength (MPa) still increased as the TiB_2 volume fraction increased. These results indicate that increasing the TiB_2 volume fraction was effective for enhancing the Fe/PA6 joint strength via the micro-scale structure fabricated from the Fe-Ti-B powder mixture. In Fig. 12(d), the lap shear strength (MPa) was plotted as a function of the product of the area fraction of the particles and the bonding ratio shown in Fig. 8(c). The lap shear strength increased almost

linearly with increasing the product of the area fraction and the bonding ratio. This trend was consistent with A5052/PA6 joints via the micro-scale structure fabricated from the Al-Ti-C powder mixture (Kim et al., 2019, 2021). A high area fraction of particle-shaped products indicates that numerous particle-shaped products contributing to mechanical interlocking are formed on the Fe substrate. A high bonding ratio indicates that the particle-shaped products are well-bonded with the substrates and withstand the lap shear force without detaching from the substrate. The correlation in Fig. 12(d) suggested that numerous particle-shaped products well-bonded with the substrate produced a strong interlocking with PA6 and enhanced the joint strength.

To check the validity of the correlation, the fracture surfaces of the Fe side were observed by SEM. Fig. 13 shows the fracture surfaces of Fe/PA6 joints via the micro-scale structures fabricated from the powder mixtures with $V_{\text{TiB}_2} =$ (a) 60 %, (b) 80 %, and (c) 100 %. The tensile direction of the PA6 was shown by the black arrow near Fig. 13(c). Most of the particle-shaped products were retained on all the specimens even after the lap shear tests. In Fig. 13(c), the particle-shaped products at the edge of the laser-scanning line, which were poorly bonded with the substrate (Fig. 7d), were detached (indicated by red arrow), validating the importance of well-bonding of the particle-shaped products with the substrate for withstanding the lap shear force. The PA6 also remained on the particle-shaped products and were located at the opposite side of its tensile direction, as indicated by yellow arrows, showing that PA6 was partially fractured at the joint interface by strongly interlocking with particle-shaped products. More PA6 was interlocked with the particle-shaped products fabricated from the powder mixture with a higher TiB_2 volume fraction due to a larger amount of retained particle-shaped products. The fracture mode was well consistent with the linear correlation between the lap shear strength and the product of the area fraction of the particle-shaped products and the bonding ratio (Fig. 12d). Thus, the numerous particle-shaped products well-bonded with the substrate enhanced mechanical interlocking and the joint strength.

The TiB_2 fraction in the particle-shaped products increased with increasing the Ti and B fractions in the powder mixture (Fig. 11). In general, the addition of TiB_2 into Fe increases Young's modulus, hardness, and strength. The particle-shaped products would be reinforced by increasing the TiB_2 volume fraction. There is a possibility that the reinforcement enhanced the joint strength. For example, the particle-shaped products might yield during the lap shear test at a low TiB_2

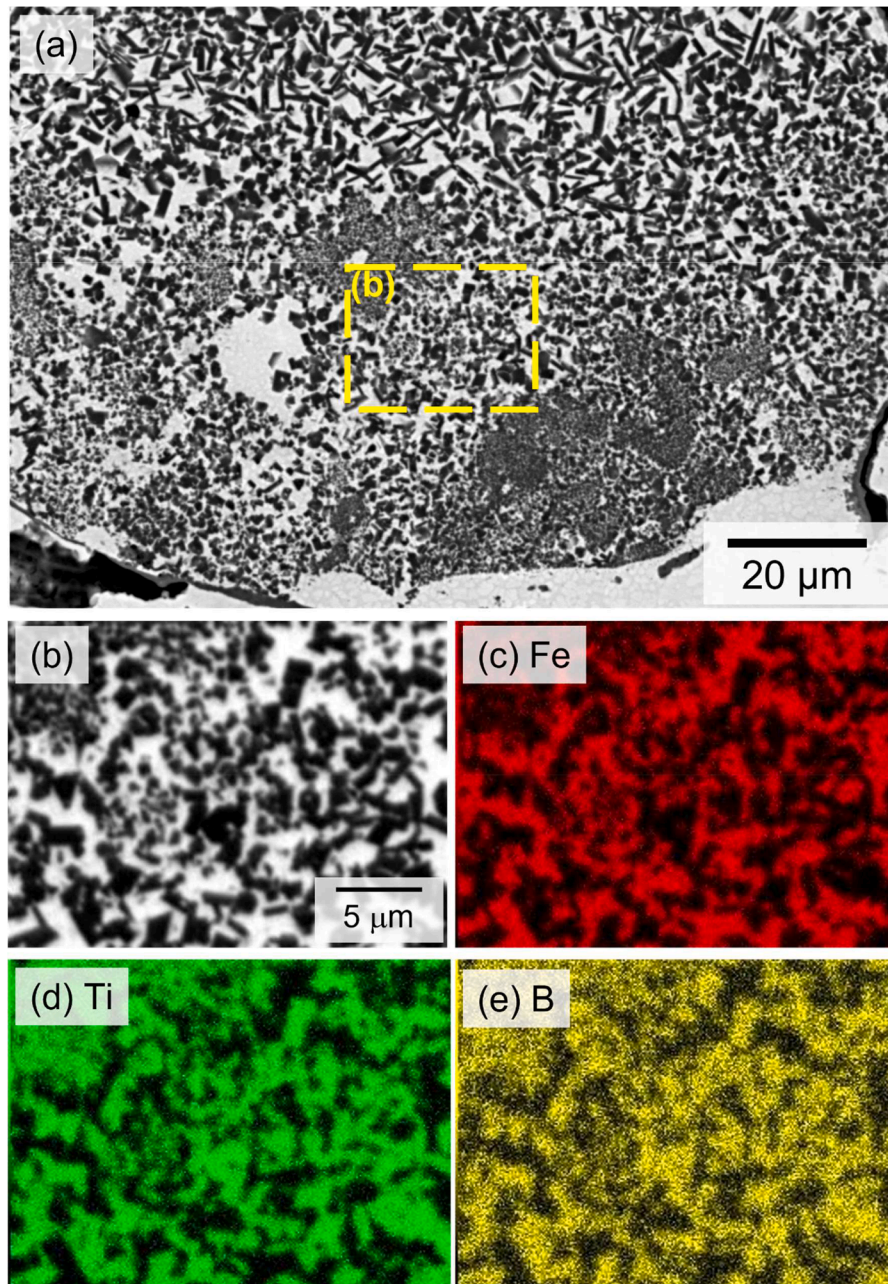


Fig. 10. (a) Low- and (b) high-magnification SEM images and the corresponding EDS element maps of (c) Fe, (b) Ti, and (c) B showing the microstructure inside the micro-scale structures in the particle-shaped products under $V_{\text{TiB}_2} = 60\%$.

volume fraction, and the yielding might be suppressed by increasing the TiB_2 volume fraction. However, an in-situ synthesized Fe/ TiB_2 composite (TiB_2 volume fraction: $\sim 13\%$) through laser powder bed fusion process exhibited a high yield stress of approximately 1400 MPa (Feng et al., 2023), which is incomparably higher than the tensile strength of PA6 (approximately 70 MPa (Güllü et al., 2006)). The yielding of the particle-shaped products during the lap shear tests would be negligible. Another possibility is the stress singularity, which needs to be considered at the interface of the joints with a large difference in Young's modulus. A large difference in Young's modulus increases stress concentration at the joint interface edge and decreases the joint strength. Young's modulus of PA6, α -Fe, and TiB_2 are 1.1 GPa, 205 GPa, and 550 GPa, respectively. An increase in the TiB_2 volume fraction enlarges the difference in Young's modulus and might produce the stress singularity. Therefore, an increment in Young's modulus would not cause an enhancement in the joint strength with the TiB_2 volume fraction. To

clarify the effect of stress singularity on the joint strength, finite element modeling will be required.

Discussion

Uniform particle-shaped products were formed at TiB_2 volume fractions exceeding 60 % (Fig. 6). The area fraction of these products and the overall micro-scale structure area increased with TiB_2 volume fraction, while the bonding ratio between the products and the substrate decreased (Fig. 8). Inside the particle-shaped products, an Fe/ TiB_2 two-phase microstructure was formed (Figs. 9–11). These structural changes contributed to the increase in lap shear strength with increasing TiB_2 volume fraction (Fig. 12). The effect of the TiB_2 volume fraction on the morphology of the micro-scale structure is discussed below.

In the previous study related to the micro-scale structure fabricated from the Al-Ti-C powder mixtures, the TiC network was formed inside

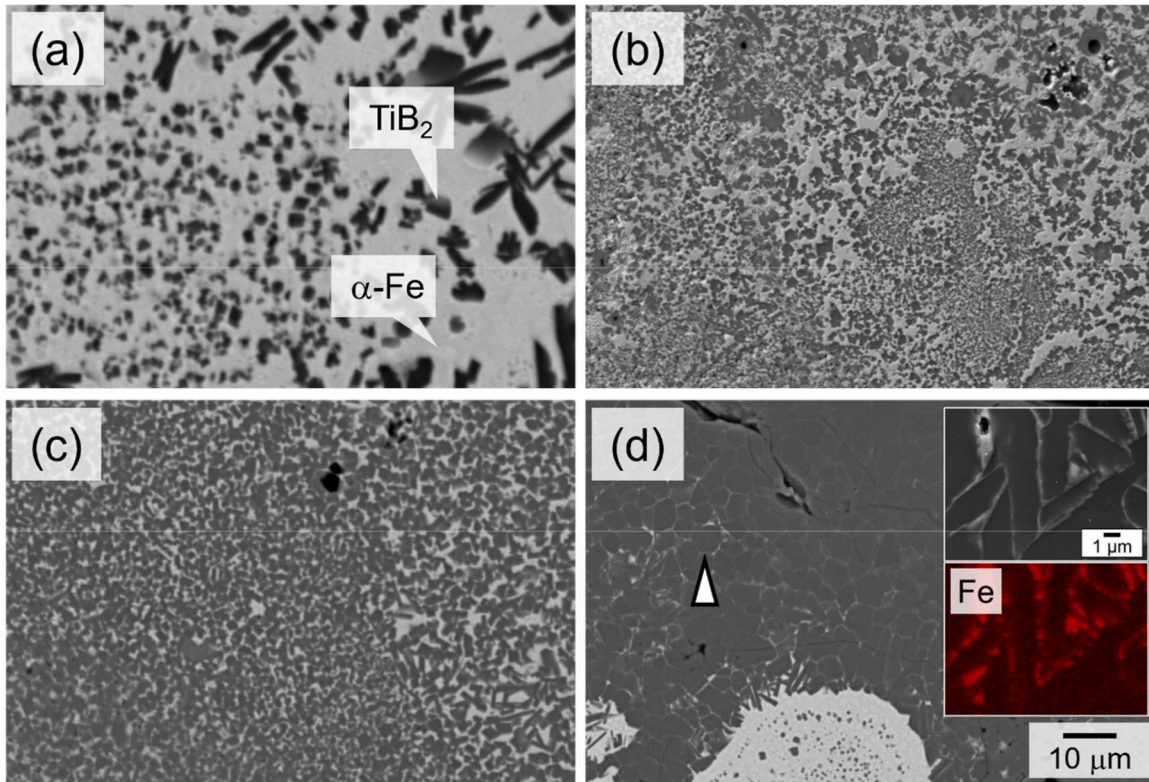


Fig. 11. High-magnification SEM images showing the microstructure inside the particle-shaped products fabricated using powder mixtures with V_{TiB_2} = (a) 40 %, (b) 60 %, (c) 80 %, and (d) 100 %.

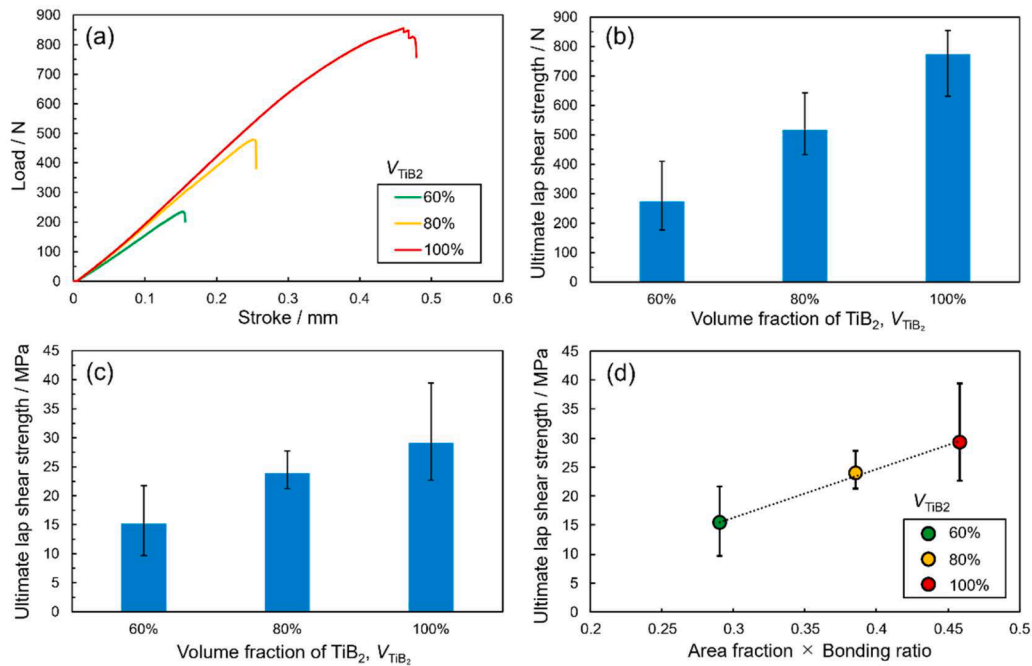


Fig. 12. (a) Load-stroke curves for the Fe/PA6 joints and quantified ultimate lap shear strength with a unit of (b) N and (c) MPa. (d) Relationship between ultimate lap shear strength (MPa) and the product of the area fraction of the particles and the bonding ratio.

the particle-shaped products and played a role in the skeleton structure to keep the shape of the particle-shaped products. The necessity of the Ti and B additions to form the particle-shaped products shown in Fig. 6 suggested the possibility of forming a TiB_2 network inside the particle-shaped products. To investigate the possibility, the Fe substrate with

particle-shaped products with $V_{TiB_2} = 60\%$ was etched using a Nital solution and observed by the SEM. The SEM images of etched particle-shaped products are shown in Fig. 14. At the low-magnification image, the particle-shaped products were locally etched (Fig. 14a). A high-magnification observation at the locally etched area showed that

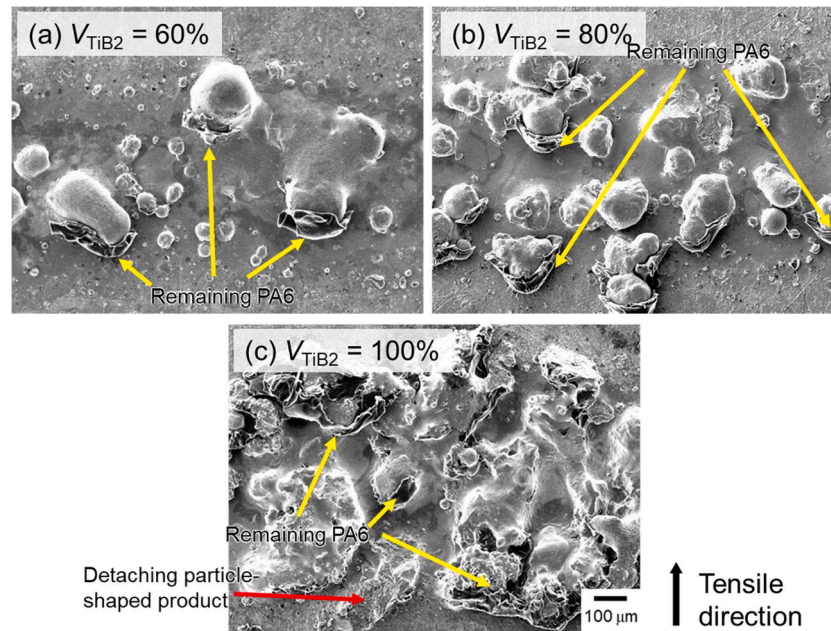


Fig. 13. SEM images showing the fracture surface of the Fe/PA6 joints via the micro-scale structures fabricated from the powder mixtures with V_{TiB_2} = (a) 60 %, (b) 80 %, and (c) 100 %.

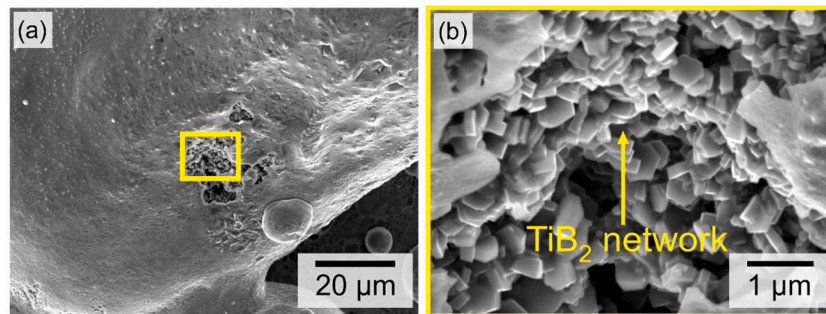


Fig. 14. (a) Low- and (b) high-magnification SEM images showing the surface of etched particle-shaped products in the micro-scale structure formed from the powder mixtures with V_{TiB_2} = 60 %.

hexagonal-shaped TiB_2 particles were interconnected with each other and formed a network structure. Therefore, the TiB_2 network likely plays a role as a skeleton to maintain the morphology of the particle-shaped products, analogous to the TiC network. An important factor in forming the TiB_2 network was the volume fraction of the TiB_2 formed inside the particle-shaped products. As shown in Fig. 11(a), the TiB_2 particles were dispersed in the particle-shaped products fabricated from the powder mixture with V_{TiB_2} = 40 % in two-dimensional cross-sections. The small TiB_2 volume fraction was less likely to form the TiB_2 network, resulting in a low area fraction of the particle-shaped products (Figs. 6a, 6b, and 8a). The densely distributed TiB_2 formed from the powder mixture with $V_{\text{TiB}_2} \geq 60$ % (Figs. 6c and 8a) was interconnected with each other, facilitating the formation of the particle-shaped products via the skeleton TiB_2 .

Another effect of increasing the TiB_2 volume fraction was the formation of particle-shaped products with reduced adhesiveness to the substrate far from the center of the laser scanning line, resulting in a decrease in bonding ratio and an extension of the micro-scale structure area. The laser used in this study exhibited a Gaussian energy distribution, which meant that the laser power input became lower at the edge of the laser scanning line. This reduced power input would insufficiently melt the Fe substrate, resulting in a low bonding ratio at the edge of the laser scanning line. However, if this mechanism dominated the

formation of particle-shaped products with low adhesiveness to the substrate far from the center of the laser scanning line, it should occur regardless of the TiB_2 volume fraction and should not extend the area of the micro-scale structure. Another possibility is the combustion reaction between Ti and B that forms TiB_2 with a large amount of exotherm (323 kJ/mol at the standard condition). In this assumption, the particle-shaped products far from the center of the laser scanning line were bonded with the substrate dominantly by the heat generated by the combustion reaction. To investigate the contribution of reaction heat, the temperature was monitored in situ by a high-speed infrared camera during laser scanning of powder mixtures with V_{TiB_2} = 60 % and 100 % and are shown in Fig. 15. As illustrated in Fig. 15(a), the laser scanned from back to front in each image. In the case of V_{TiB_2} = 60 %, the high-temperature region moved along the laser scanning line. In contrast, when V_{TiB_2} was 100 %, the high-temperature region moved not only in the laser scanning direction but also in the opposite direction when V_{TiB_2} was 100 %. Furthermore, the high-temperature region in the case of V_{TiB_2} = 100 % was obviously wider than that of V_{TiB_2} = 60 %. These results indicated that the combustion reaction heat, as well as the laser energy, contributed to raising the temperature at the micro-scale structures and would affect its morphology. Consequently, the slightly reduced bonding ratio due to poor adhesiveness far from the laser scanning line and the extended area of the micro-scale structure can be

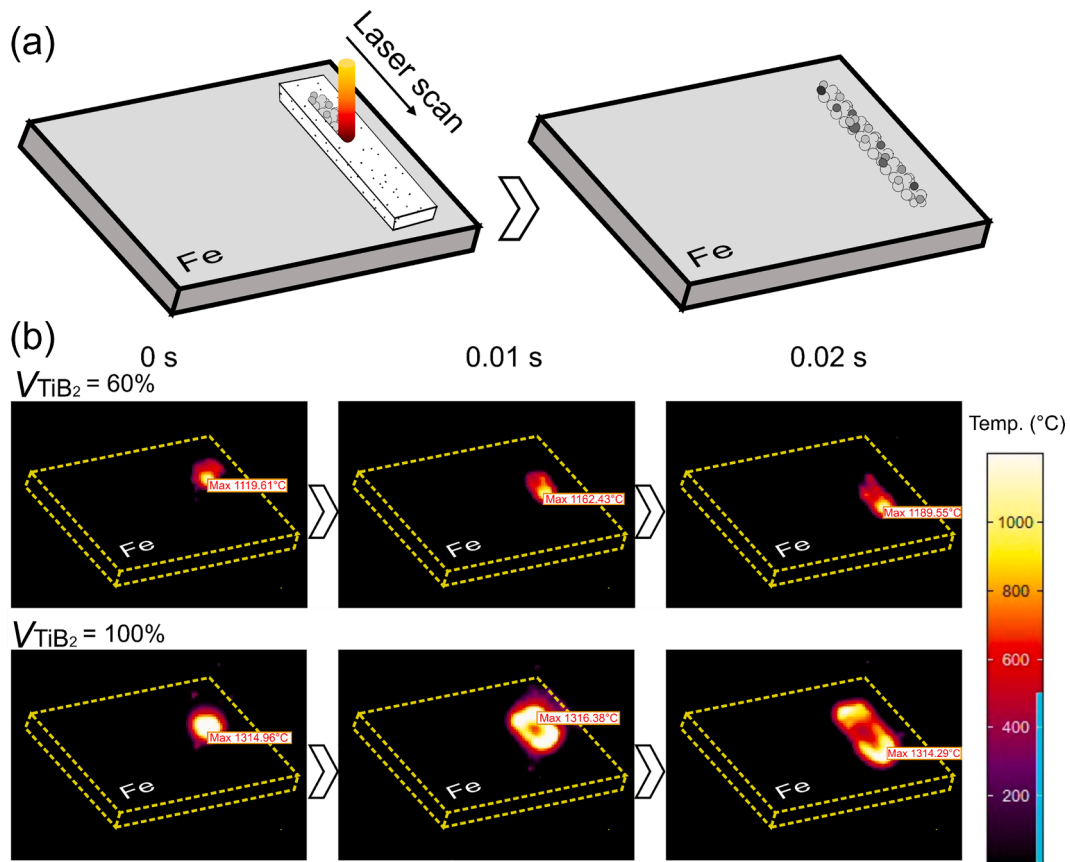


Fig. 15. (a) Schematic illustration showing the laser scan direction in the infrared (IR) images and (b) temperatures measured by the IR camera during the laser-scanning to the powder mixtures with TiB_2 volume fraction of 60 % and 100 % at 0 s, 0.01 s, and 0.02 s.

attributed to the combustion reaction between Ti and B induced by the laser scanning.

Conclusion

This study attempted to fabricate a micro-scale structure (consisting of particle-shaped products) on a pure Fe substrate via laser-scanning to Fe-Ti-B system powder mixture for joining with polymer. The effects of powder composition (TiB_2 volume fraction) on the morphology of the micro-scale structure, microstructure, and joint strength with PA6 were investigated. The following conclusions were obtained:

- The addition of Ti and B was indispensable for forming the particle-shaped products for mechanically interlocking with polymer. The particle-shaped products were composed of Fe/ TiB_2 composites. The volume fraction above 60 % was required to form uniformly dispersed particle-shaped products. The threshold TiB_2 volume fraction would be attributed to the formation of the TiB_2 network.
- The area fraction of particle-shaped products increased with increasing the TiB_2 volume fraction. Higher TiB_2 volume fractions led to the formation of an interconnected TiB_2 network, which served as a skeletal structure for the particle-shaped products.
- A higher TiB_2 volume fraction generated the particle-shaped products with poor adhesiveness to the substrate far from the center of the laser scanning line, leading to a slightly lower bonding ratio (an indicator of adhesiveness) and a larger area of micro-scale structure. This would be caused by the contribution of the combustion reaction during the laser scanning to the bonding of the particle-shaped products at the region far from the

center of the laser scanning line where the laser energy input was low.

- Fe/PA6 joint strength (with units of N and MPa) increased consistently with higher TiB_2 volume fractions. The joint strength was linearly correlated with the product of the area fraction of the particles and the bonding ratio. Despite the slight degradation in adhesiveness of the particle-shaped products to the substrate at higher TiB_2 volume fractions, the enhanced formation of particle-shaped products contributed to the high joint strength. In this study, the highest joint strength of approximately 30 MPa was achieved at $V_{TiB_2} = 100\%$.

As a concluding remark, the micro-scale structure for a high-strength Fe/PA6 joint was achieved in this study. Various advanced joining technologies including ultrasonic welding (Yeh et al., 2016), friction-assisted welding (Lambiase et al., 2017), and laser-assisted welding (Ye et al., 2021) have been developed for efficiently achieving metal/polymer or metal/carbon fiber reinforced thermoplastic (CFRTP) joining. The micro-scale structure will help the joinability (Iwata et al., 2022) and enhance the strength of Fe/polymer and Fe/CFRTP joints fabricated through these processes. A combination of micro-scale structuring and advanced joining technologies needs to be studied in the future.

CRediT authorship contribution statement

Shaoyun Zhou: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Koki Omiya:** Methodology, Investigation, Formal analysis, Data curation. **Yuto Ueda:** Methodology, Investigation, Formal analysis, Data curation. **Asuka Suzuki:** Writing – review & editing, Supervision, Resources, Project administration,

Funding acquisition. **Naoki Takata**: Writing – review & editing, Supervision, Funding acquisition. **Makoto Kobashi**: Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the “Knowledge Hub Aichi” Priority Research Project of the Aichi Prefectural Government (Japan) and Hitachi Metals and Materials Science Foundation.

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